ESTARTA COMPANY

ANNUAL REPORT

EMPLOYEES DATA ANLAYSIS

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AGENDA

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01
INTRODUCTION

Data analysis plays an important role in understanding organization and company. This report analyzes employee data in order to explore the characteristics of the data and to understand the relationship between each column with the other, such as employee distribution across departments , salary and important information.
The goal of this analysis is to understand employee distribution, salary pattern and characteristics within the company. The employee dataset contains various attributes including department, age, salary, bonus, country, ethnicity,hire date ,city. Through the data visualization techniques using matplotlib and pandas will present clear insight into the company structure.

02
OVERVIEW

The dataset used in this analysis contains detailed information about employees working within a company. It includes several attributes related to employee demographics, job roles, and compensation. Examples of the variables included in the dataset are employee name, job title, department, gender, age, annual salary, bonus percentage, as well as geographic information such as country and city.

These variables provide a comprehensive overview of the workforce structure and allow for a better understanding of how employees are distributed across different departments and roles within the organization. By examining these attributes, it becomes possible to analyze workforce characteristics, salary distribution, and demographic patterns within the company.

Analyzing this dataset helps identify patterns related to employee distribution, compensation levels, and organizational structure.

03

DATA CLEANING

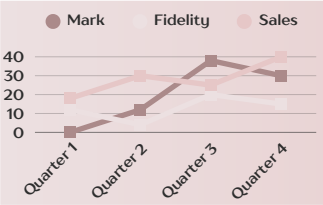
Before performing the analysis, the dataset was carefully cleaned to ensure the accuracy and quality of the data. Data cleaning is an important step in data analysis because missing or inconsistent values may affect the reliability of the results.

Drop Column

First, the dataset was explored using Python and the Pandas library in order to understand the structure of the data and identify any missing values. During this exploration, it was observed that the “Exit Date” column contained a large number of missing values.. During this exploration, it was observed that the “Exit Date” column contained a large number of missing values.

Drop Null Data

After removing the column, the remaining dataset was examined again to check for additional missing values. It was found that the other columns contained only a small number of missing values. These remaining null values were removed from the dataset to maintain consistency and ensure better data quality.



As a result, the cleaned dataset became ready for further analysis and data visualization.

04

DATA VISUALIZATION

Data visualization is an important step in data analysis as it helps transform raw data into graphical representations that are easier to understand and interpret. By using charts and graphs, patterns, trends, and relationships within the data can be identified more clearly.

- Department Distribution
- Age Distribution
- Salary by Department
- Gender Distribution
- Age vs Salary
- Bonus vs salary
- Employee by country

Department Distribution

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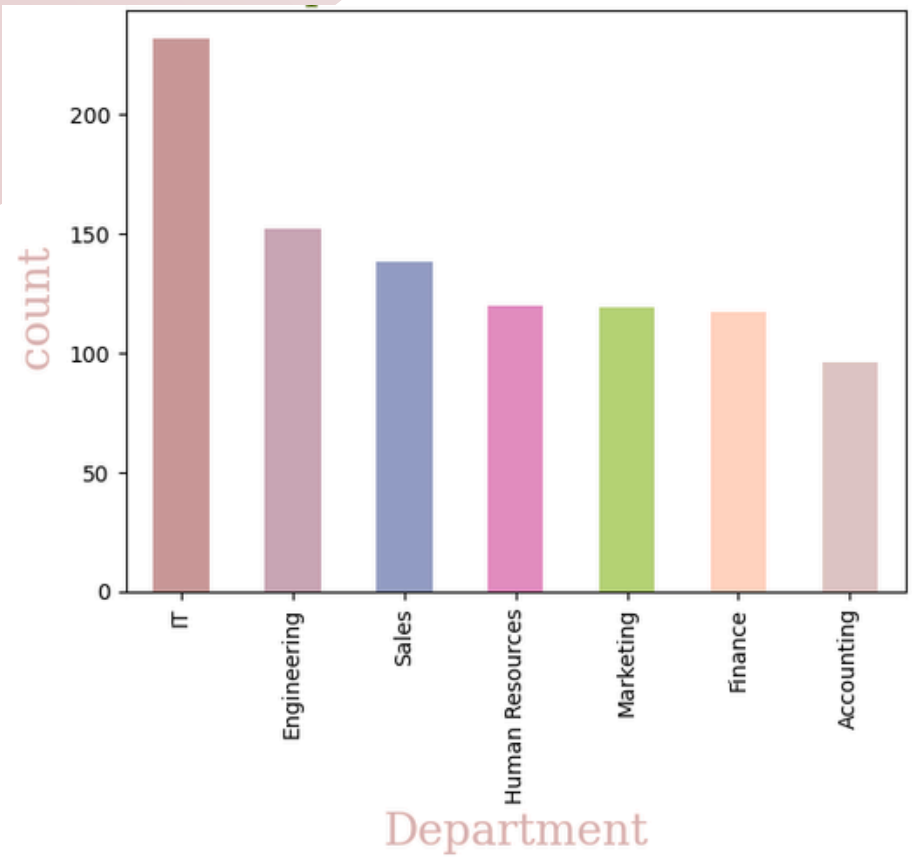

This bar chart illustrates the distribution of employees across different departments in the company. Each bar represents a department, while the vertical axis shows the number of employees working in that department.
From the visualization, it is clear that the IT department has the highest number of employees, indicating that it is the largest department in the company. This may suggest that the organization relies heavily on technology-related roles and technical staff.

The Engineering department also shows a relatively high number of employees, which may indicate a strong focus on product development or technical operations. Departments such as Sales, Human Resources, Marketing, and Finance have a moderate number of employees, suggesting balanced workforce distribution across business functions.
On the other hand, the Accounting department has the smallest number of employees, which may reflect the nature of accounting tasks that typically require fewer staff compared to operational or technical departments.

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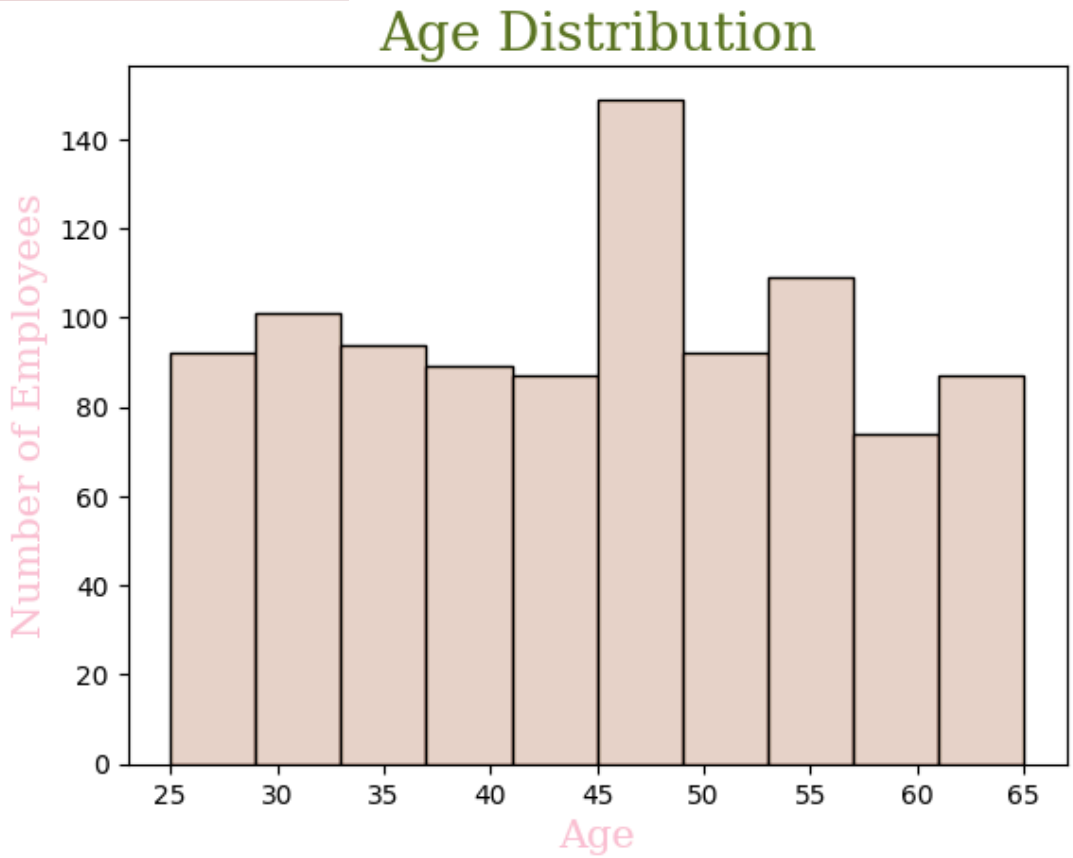
Age Distribution



This histogram shows the distribution of employee ages in the company. The horizontal axis represents the age of employees, while the vertical axis represents the number of employees within each age range.

From the chart, it can be observed that employees are distributed across a wide range of ages, starting from approximately 25 years old to around 65 years old. The distribution indicates that the company employs individuals from different age groups, suggesting a diverse workforce in terms of age.

Most employees appear to fall within the middle age ranges, particularly between approximately 30 and 55 years old. This suggests that a large portion of the workforce consists of experienced professionals who are likely to have several years of work experience.

Salary by Department

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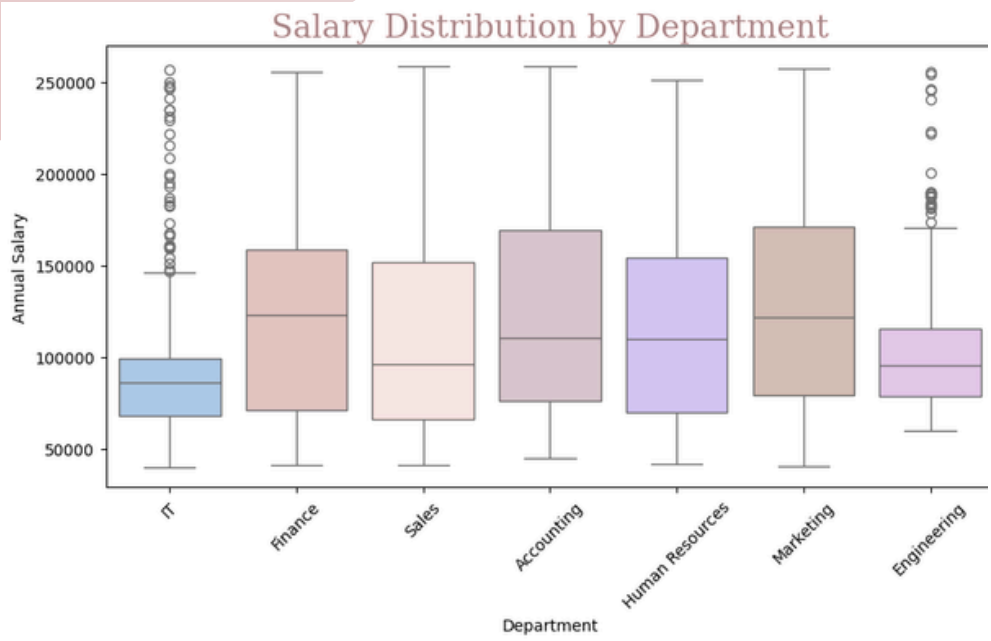

This box plot shows how annual salaries are distributed across different departments in the company. Each box represents the salary range within a department, showing the median salary and the spread of salary values.

From the chart, it can be observed that salary levels vary between departments. Some departments show higher salary ranges and greater variation in salaries, while others have more concentrated salary distributions.

Gender Distribution

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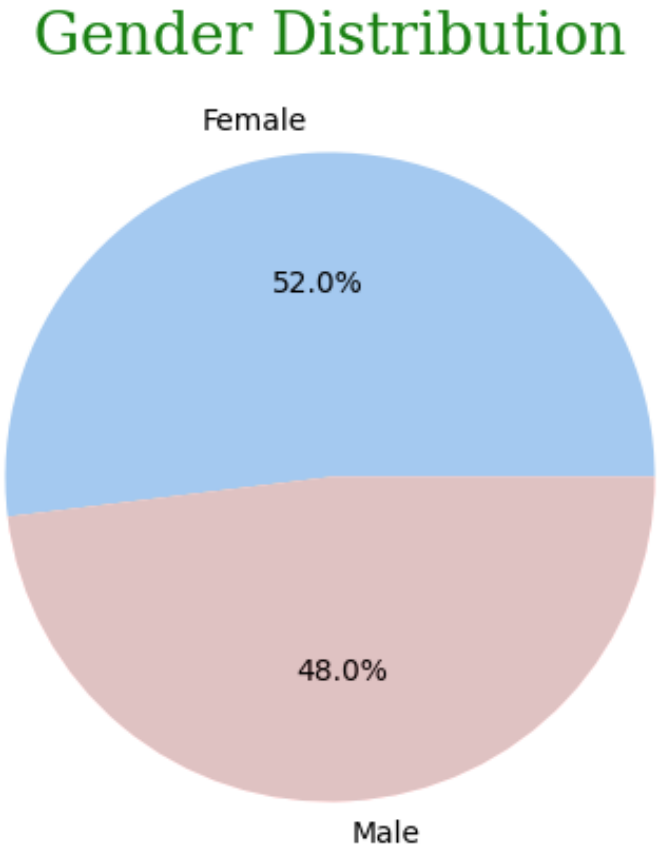

This pie chart shows the distribution of employees based on gender in the company. The chart indicates that female employees represent approximately 52% of the workforce, while male employees represent about 48%.
The distribution appears relatively balanced, suggesting that the company maintains a fairly equal representation of male and female employees within the organization.

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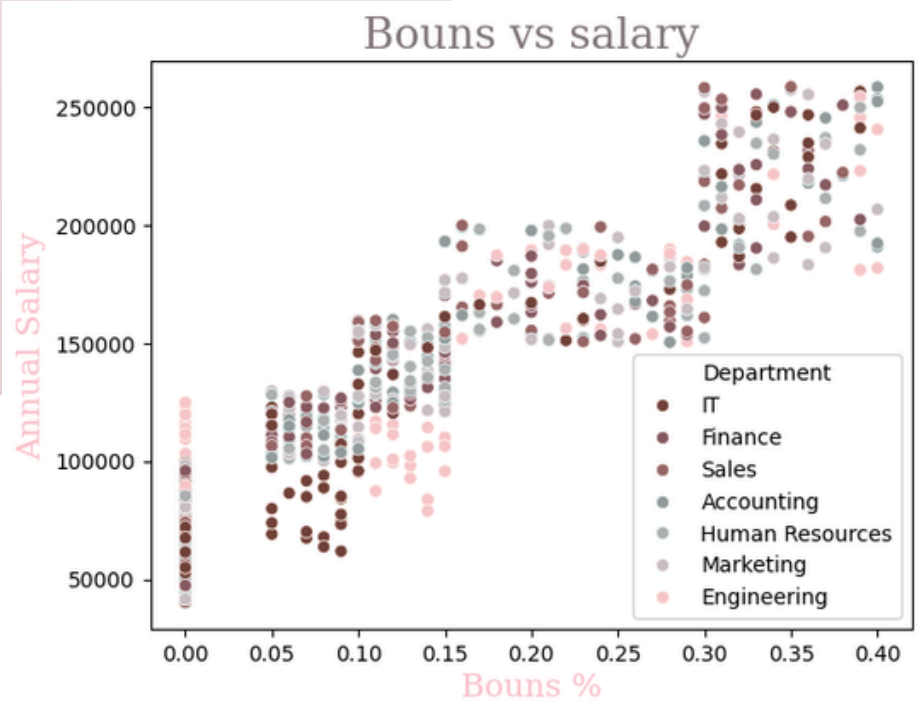
Bonus vs salary



Each point represents an employee, where the horizontal axis shows the bonus percentage and the vertical axis shows the annual salary.

From the chart, it appears that employees with higher bonus percentages tend to have higher annual salaries. This suggests a positive relationship between bonus percentage and salary levels.

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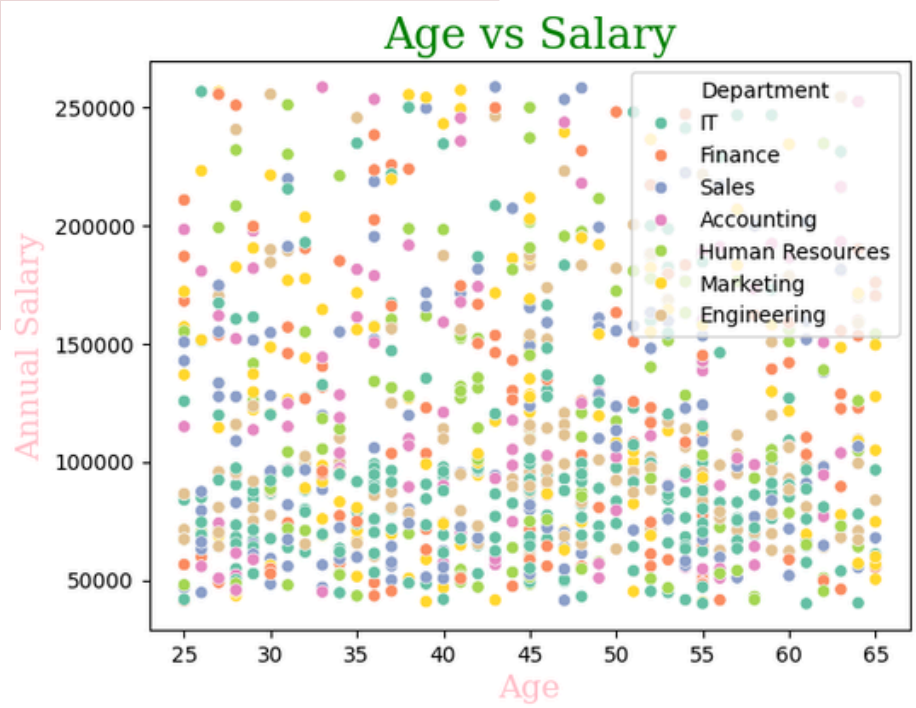
Age vs Salary



This scatter plot shows the relationship between employee age and annual salary. Each point represents an employee, where the horizontal axis shows the age and the vertical axis shows the annual salary.

From the chart, salaries vary across all age groups, and there is no clear linear relationship between age and salary. However, it can be observed that employees of different ages receive a wide range of salaries depending on their roles, experience, and department.

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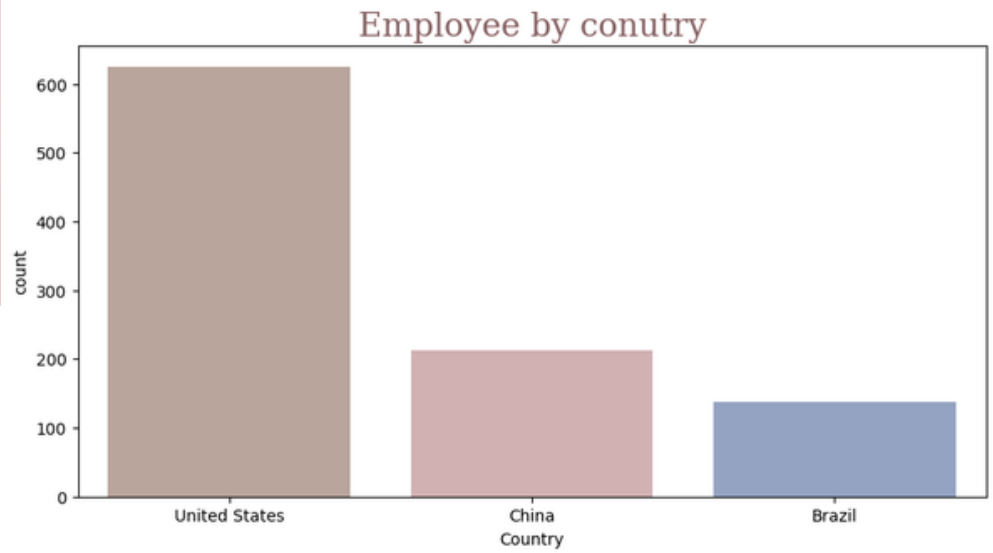
Employee by country



the countries, while the vertical axis represents the number of employees in each country.
From the chart, it can be observed that the United States has the highest number of employees, followed by China, while Brazil has the smallest number of employees. This indicates that the majority of the company's workforce is located in the United States.

05
CONCLUSIONS

This report analyzed employee data to understand workforce distribution and salary patterns within the company. The visualization showed how employees are distributed across departments, age group, and geographic location.

The analysis also highlighted differences in salary levels between departments and showed relationships between variables such as salary, bonus percentage, and employee age.

Overall, data visualization helped provide clear insights into the company's workforce structure and compensation patterns

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THANK YOU

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